

UPSC Prelims

Science & Tech

Class No. 8

Space Technology

Space Technology



- Weightage : 1 or maximum 2 questions in prelims
- So we will study as per its importance
- Today : let's do basics of space tech

Gravitational Force



- Gravity is bending of space time
- However, we will understand it like a force for now.

$$F_{Gravity} = G \frac{M1 \times M2}{R^2}$$

- G = Gravitational Constant = $6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
- As you can see, this is a very small number
- So for gravity to operate, you need a very large mass
- Otherwise, Gravitational force will be extremely weak.

Orbits

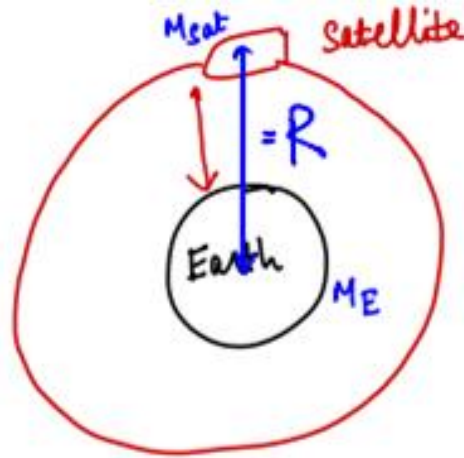


- Definition
- Classification on the basis of :
 - shape
 - revolution
- Distance from earth

Orbits



Imaginary path taken by a celestial body or spacecraft centripetal force due to gravitational force is the driving force behind the orbit.



Classification of Orbits



1. On basis of shape

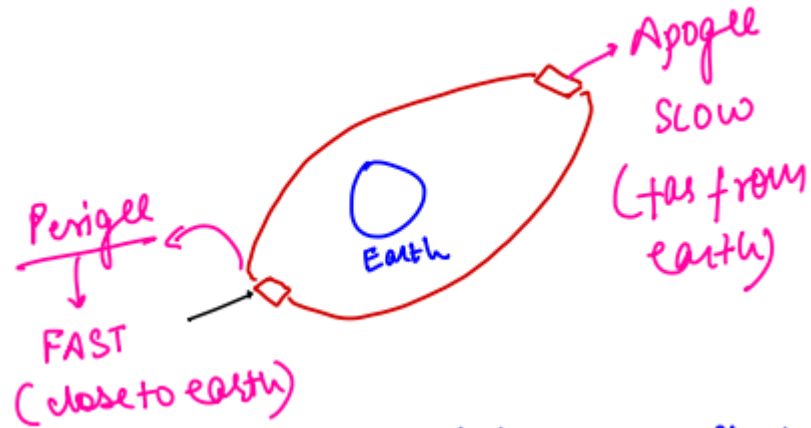
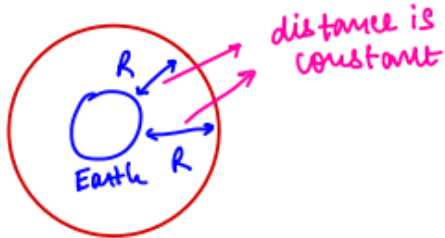
(a) Elliptical Orbit : orbit of planets

(b) Circular Orbit : eg: Geostationary

Classification of Orbits



Mv^2/R is constant



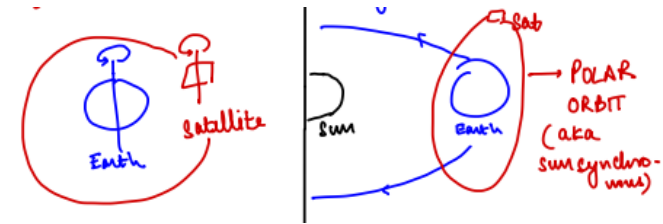


ELLIPTICAL	CIRCULAR
Distance variable speed variable velocity variable	Distance constant speed constant velocity variable (as direction is changing)

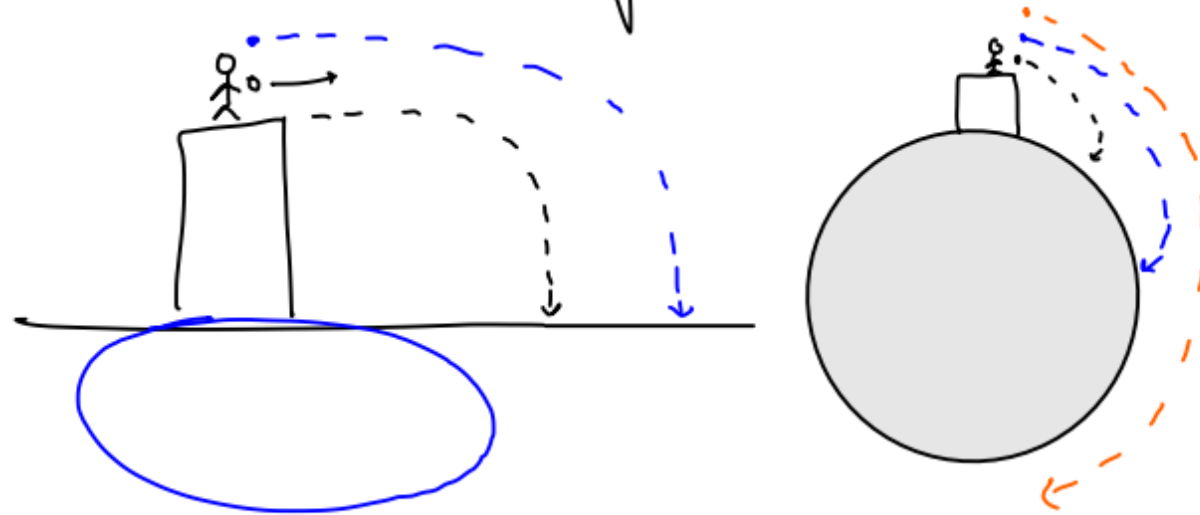
Classification : On Basis of Revolution



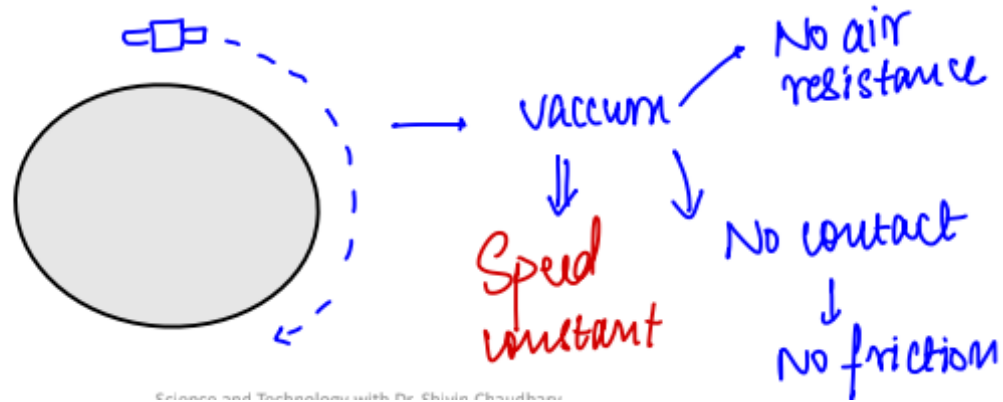
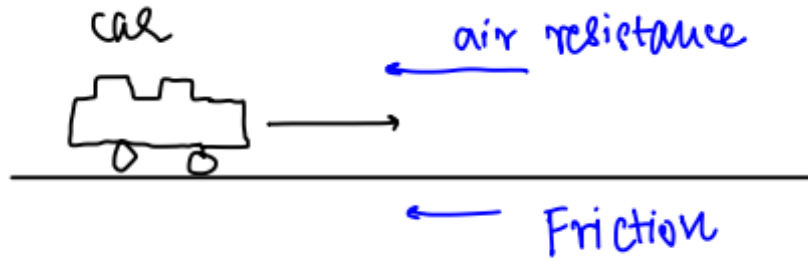
Geo Synchronous	Sun synchronous
(i) Satellite's revolutions is synchronised as per earth's Rotation (ii) Orbits in equatorial plane (iii) Application: (a) Communication (b) Navigation	(i) Satellites revolution is synchronised on basis of Earth's Revolution (ii) Orbits in Polar plane - Application (a) Earth Observation (b) Remote sensing



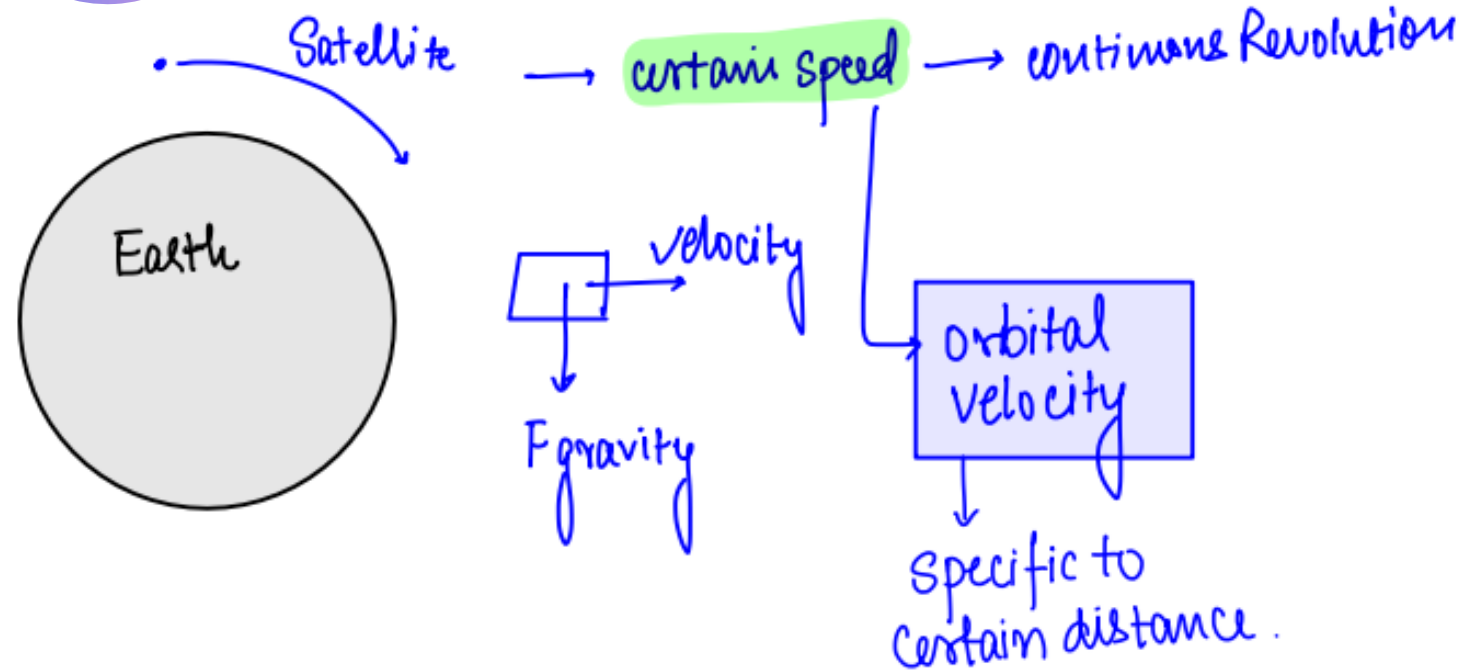
Why Satellites don't fall



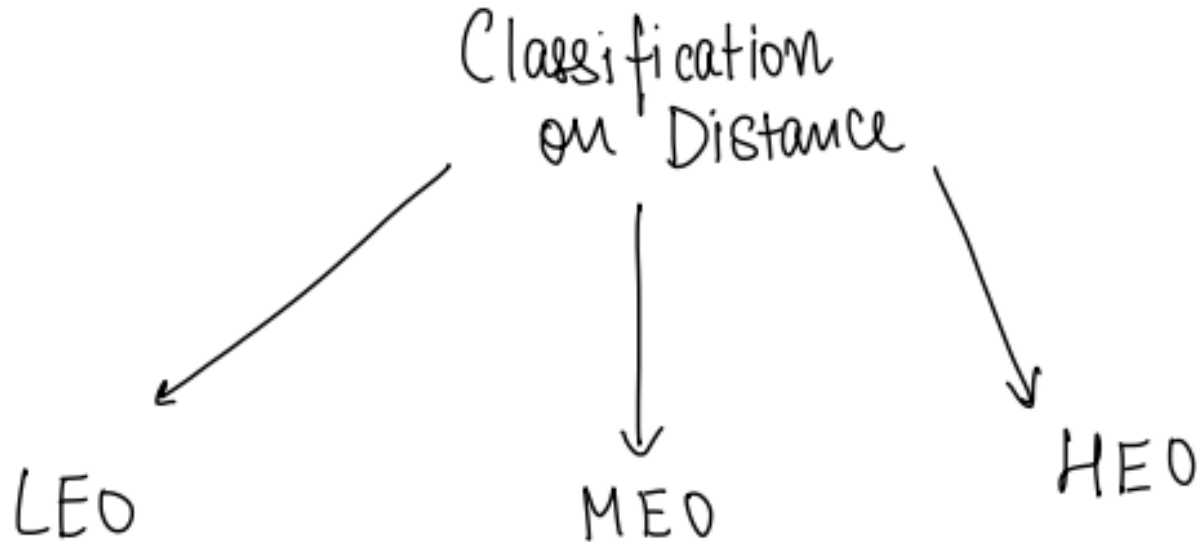
at certain speed → overcomes curvature



Science and Technology with Dr. Shivin Chaudhary



Orbital velocity is specific to a certain distance.



Low Earth Orbit



- O- 2000 km from earth's surface.
- Near earth orbit : 400-800 km
- Applications -
 - (i) Earth observation (Remote sensing)
 - (ii) Astronomical observation (Telescope) **why in LEO?**

Low Earth Orbit



- Astronomical observation : Easy servicing in case of defaults
- (ii) LEO allows to capture data such as radiation (cosmic rays, x rays, gamma rays)
 - Eg: Hubble, Kepler - LEO
 - Satellites : eg: Astrosat 650 Km (by India)

Hubble

- Launch in 1990
- Problem in Lens - NASA sent astronauts on space walk

Application Continued:



(iii) Space Station is conducted is LEO

Eg : International Space Station at - 384 km

by : NASA (USA) , Roskosman (Russia), ESA (Europe), JEXA (Japan), CSA (Canada)

- Tiangong - China (Independent)

Concerns : Weaponization, Spying, Control and Monopoly

Can you relate? Countries of ISS and global north



Applications continued : Space Stations



- (a) Microgravity experimentation (to test products at zero gravity)
 - Eg. Metallurgy, artificial photosynthesis
- b) Space observation
- c) Space Voyage - acts like terminal
- d) servicing workshop
- e) Space Tourism

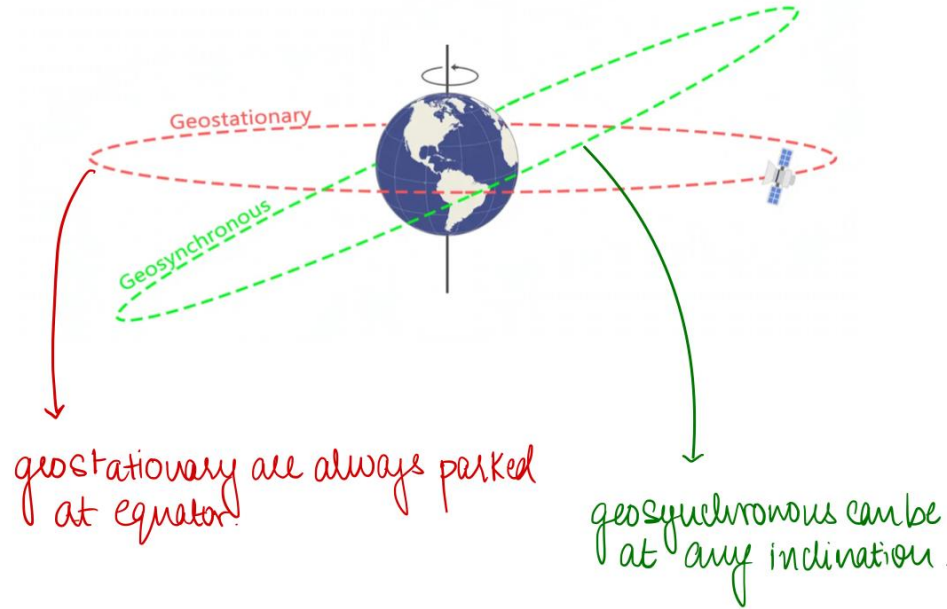
Medium Earth Orbit



- MEO : from 2000 km to 36000 km.
- Includes geostationary and geosynchronous orbits.

Geostationary v/s Geosynchronous.

→ Both are at 35,786 km above surface.



Centripetal force is provided by gravitational force

$$\frac{GM_e m_s}{(R + h)^2} = m_s (R + h) \omega^2$$

$$\frac{GM_e}{\omega^2} = (R + h)^3$$

$$\left(\frac{GM_e}{\omega^2} \right)^{\frac{1}{3}} = R + h$$

$$h = -R + \left(\frac{GM_e}{\omega^2} \right)^{\frac{1}{3}}$$

$$h = -6.4 \times 10^6 + \left(\frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{\frac{4\pi^2}{T^2}} \right)^{\frac{1}{3}}$$

$T = 24$ hours for geostationary satellite

$= 24 \times 60 \times 60$ second

$$h = -6.4 \times 10^6 + 42.31 \times 10^6$$

$$= 35.911 \times 10^6 \text{ m}$$

$$\approx 36000 \text{ km}$$

Let's See A video and a demonstration



- Demonstration : Geostationary orbit and Geosynchronous orbit
- Video : Orbit from earth's perspective.

MEO : Navigation Systems



- GPS (global positioning system of USA)
- GLONASS (Global Navigation and satellite system of Russia)
- GALILEO (European union)
- BEIDOU -1 (China's regional navigation in central asia)
- BEIDOU - 2 (China's global navigation system)
- NAVIC - IRNSS (India's Regional positioning system)
- QZSS (Regional System of Japan)

India: NAVIC (Navigation Indian Constellation)



- 4 Satellites (Geosynchronous)
- 3 Satellites (Geostationary 36,000 km)
- Coverage: India + 1500 Km from Border.
- Resolution (10 meters).
- All satellites have been placed.

NAVIC (NAVIGATION WITH INDIAN CONSTELLATION)



- Centre is pushing smartphone makers to enable support for its NavIC navigation system in new devices from next year
- Independent stand-alone navigation satellite system Forthcoming change: To add the L1 Spectrum band into NavIC which is part of GPS and is the most used for civilian navigational use.

NAVIC (NAVIGATION WITH INDIAN CONSTELLATION)



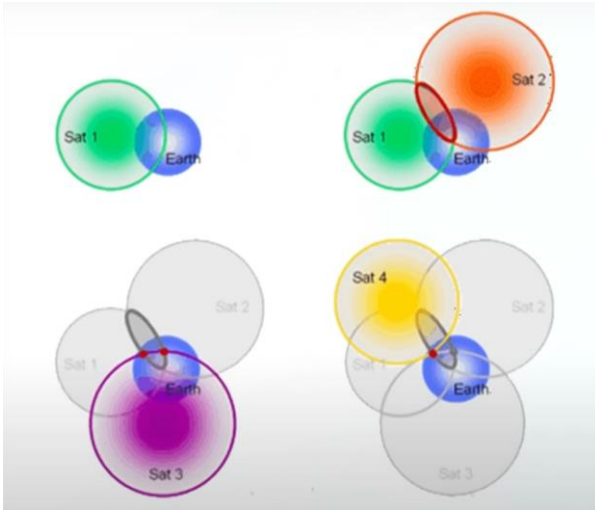
- Expansion will help to increase its use in civilian sector and ships, aircraft travelling far from the country's borders.
- Application: Public vehicle tracking in India, providing emergency warning alerts to fishermen, tracking, and providing information related to natural disasters.

GPS : NAVSTAR of USA

- Satellites are in semi synchronous orbit : half of geosynchronous.
- GPS has 24 satellites
- $R = 20,200$ km above surface
- It takes 12 hours to complete orbit

How does GPS work?

- Concept of Trilateralisation through signal testing and comparing time delays



Application of GPS systems

1. Location Services

Standard Positioning	Restricted Positioning
Civilians	Military (Use of NAVIK)
	In Kargil war, america denied GPS services to us, therefore a used of self reliance was felt.

Application of GPS systems

2. Timing Service

- Corroborates Local time.
- Use of Atomic Clocks

Application of MEO Continued:

Geo Stationary (Communication)

- (a) Circular orbit (35786)
- (b) 36000 km/22222 miles
- (c) 24 hours : time (Rotation and Revolution) : Speed of satellite matches that of earth.
- (d) Direction : West to East

High Earth Orbit

- Orbit more than 36,000 km.
- Applications : Lagrange points : lets see a video



High Earth Orbit

- L1 = 16 Lakh km from surface (Moon = 3.84 lakh km)
- L2 = Same distance otherside.
- Satellite at this point can constantly monitor the sun
- No impact of solar eclipse.

High Earth Orbit

- There are 5 Lagrange points in outer space. Here, The orbit, gravity of sun and the gravity of earth are in balance.
- If spacecraft is placed on such points, it will remain in a HALO orbit and Earth and the spacecraft together will go round the sun.
- L1, L2, and L3 points are in unstable equilibrium whereas L4 and L5 points are in stable equilibrium.
- L4 and L5 are extremely vulnerable to asteroids. Therefore even though they have stable equilibrium, L5 is preferred.

Equilibrium types

Stable



Unstable



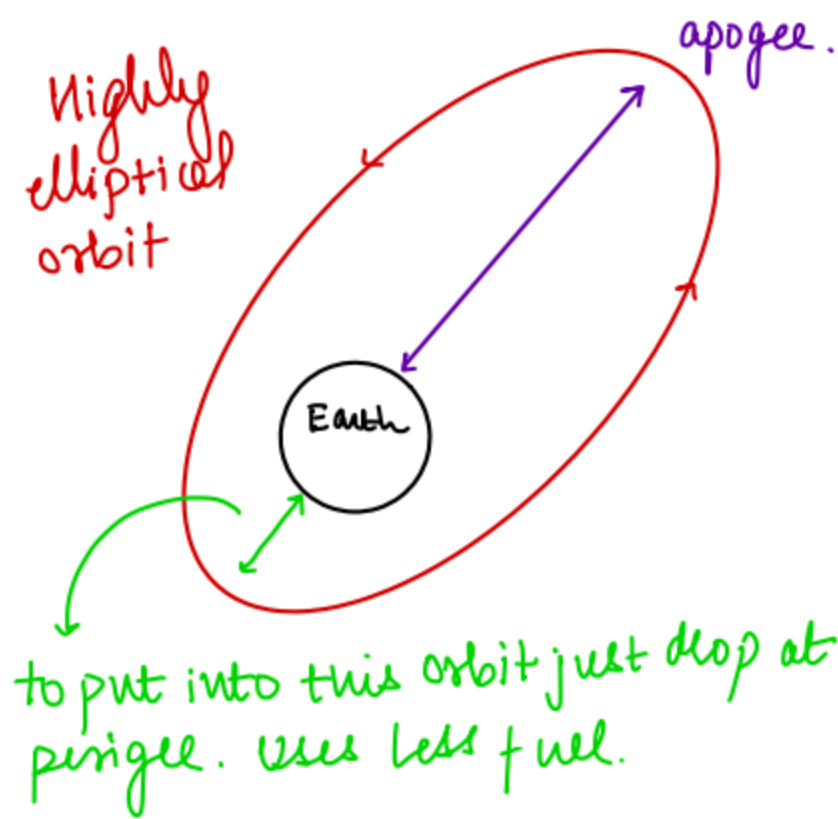
Neutral



Application of HEO: Continued.

Geosynchronous transfer orbit (GTO)

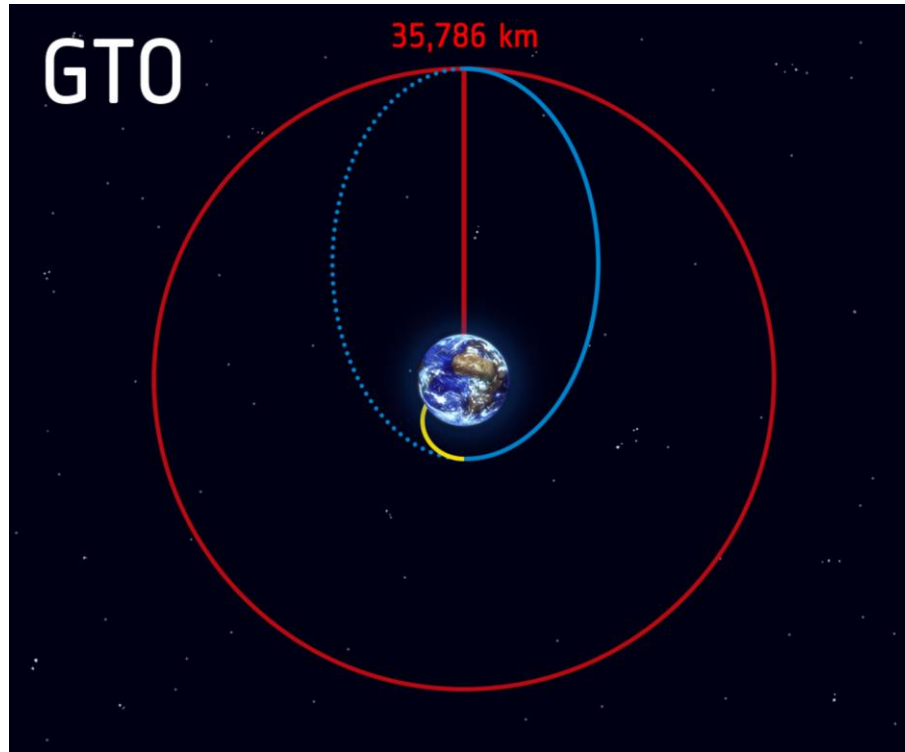
- A transfer orbit is HEO, geosynchronous with perigee = 300-400 km and apogee = 36000 km to 38000 km.
- This orbit is used for initial launch of a communication satellite or a spacecraft.
- This orbit is used by space agencies to manoeuvre a satellite for various applications.



Application of HEO: Continued.

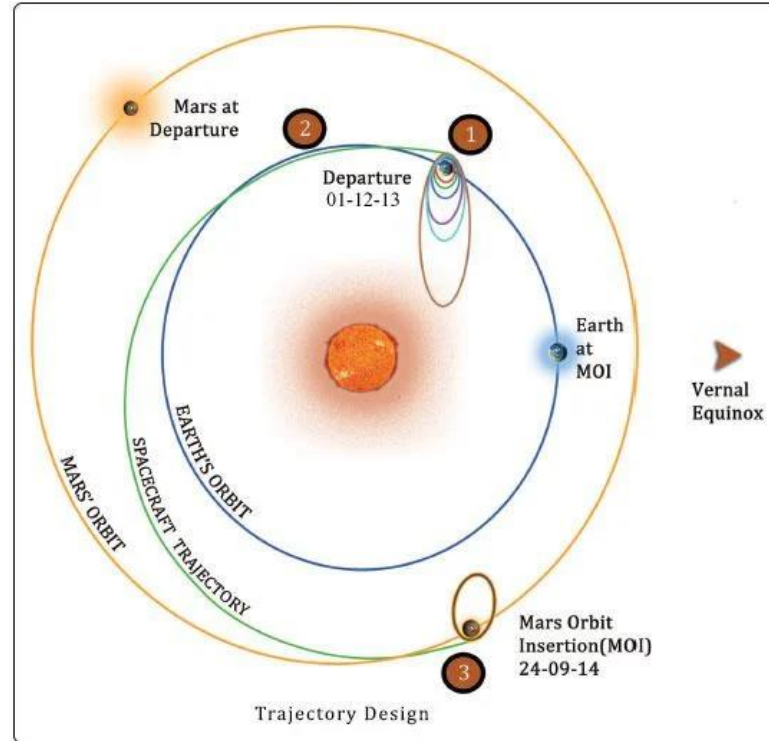
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ISRO used geosynchronous transfer orbit in Mangalyaan Mission : Cost came to be close to ₹7 / Km only.

However : there was a risk : going from heliocentric to mars orbit was the most crucial step.



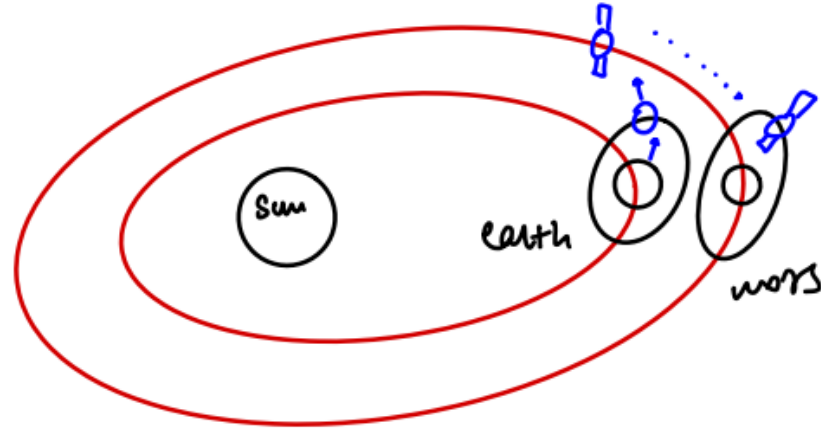
However, Certain risks are present

Geocentric - Heliocentric

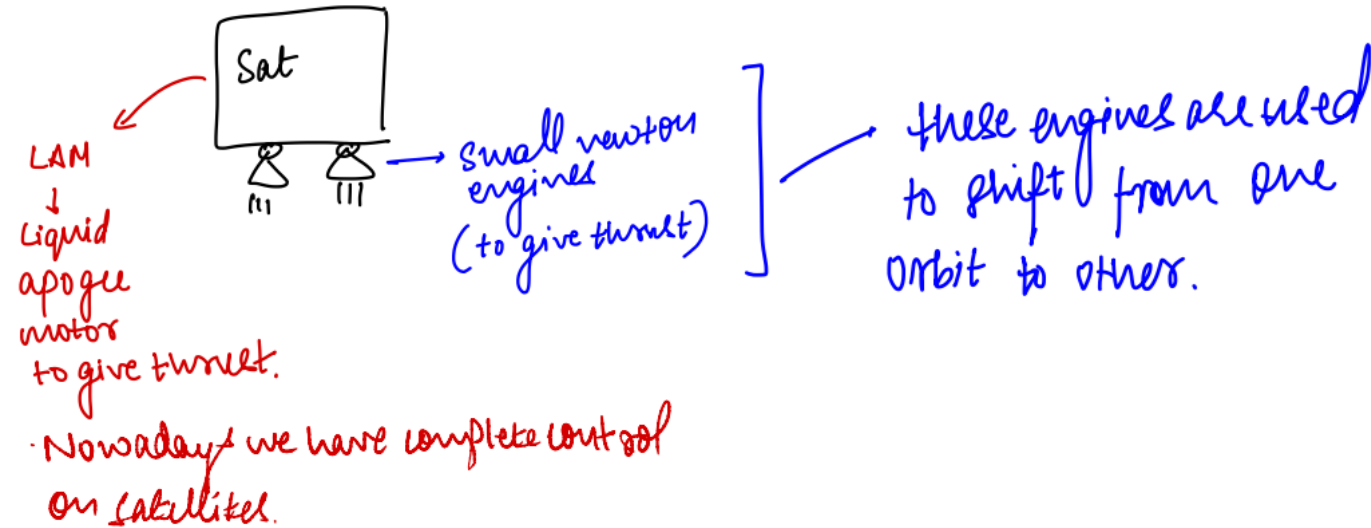
Issue - Shifting of orbits

NASA - Does not use GTO often

ISRO - Save cost



Shifting Orbit



LAUNCH VEHICLES

- Space vehicles to carry spacecraft or humans to outer space. (aka rocket)

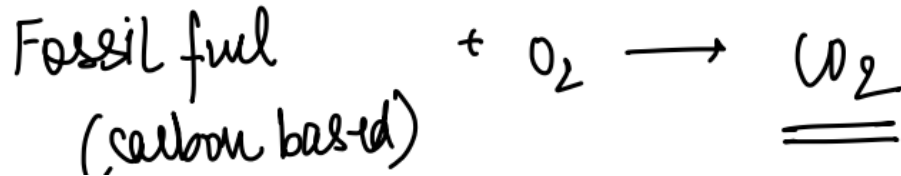
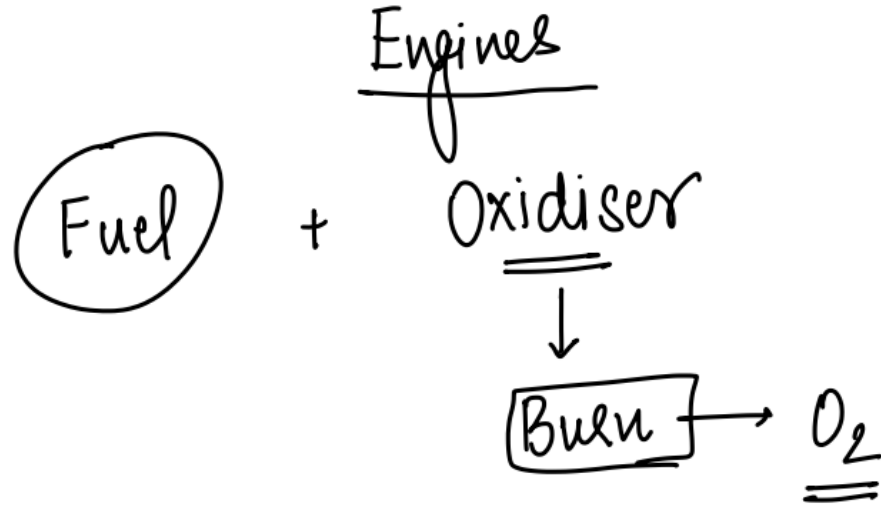
Types	
ELV (Expendable launch Vehicle)	Reusable launch vehicle
Used only once for a space mission Eg: PSLV GSLV	Can be used multiple times



ELV	RLV
● Space debris issues ● Used for Unmanned missions ● Indian only has ELV, working to develop RLV	● Returns back to earth ● Can be used for both manned and unmanned mission ● Advantage is that it is reusable and saves lot of money

Major Components of Launch Vehicles

- Engines
- Fuel
- Satellite / Payload
- Command and Control System



Components : Engines

Engines

Non air breathing propulsion system

Air breathing propulsion system (Ramjet/Scramjet)

Use of Oxidiser to burn fuel that is kept on the rocket

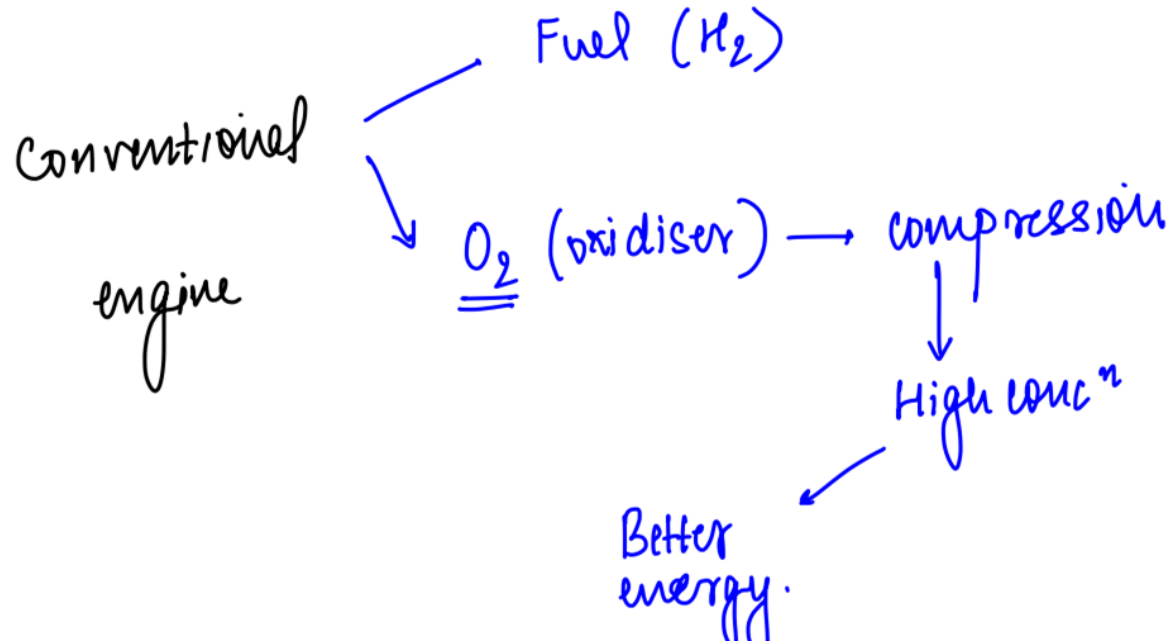
Air breathing : word breakdown

Oxidiser adds to mass of rocket

Oxygen from air to burn fuel

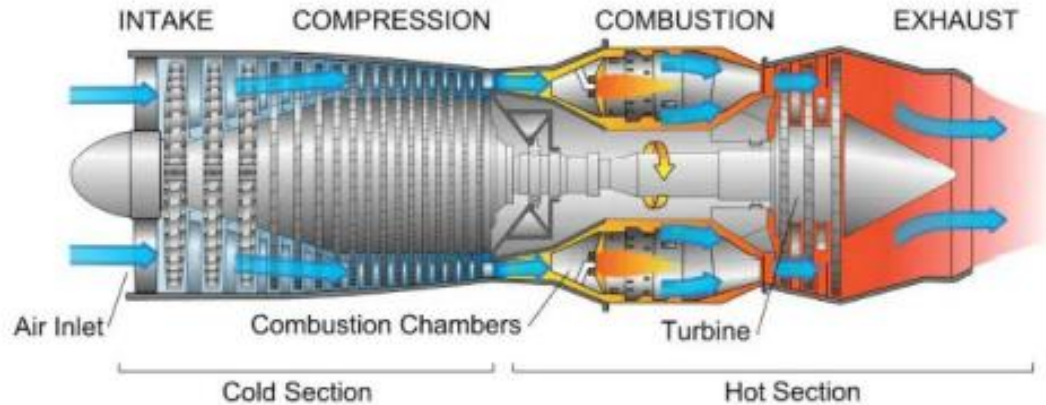
Heavier

Lighter



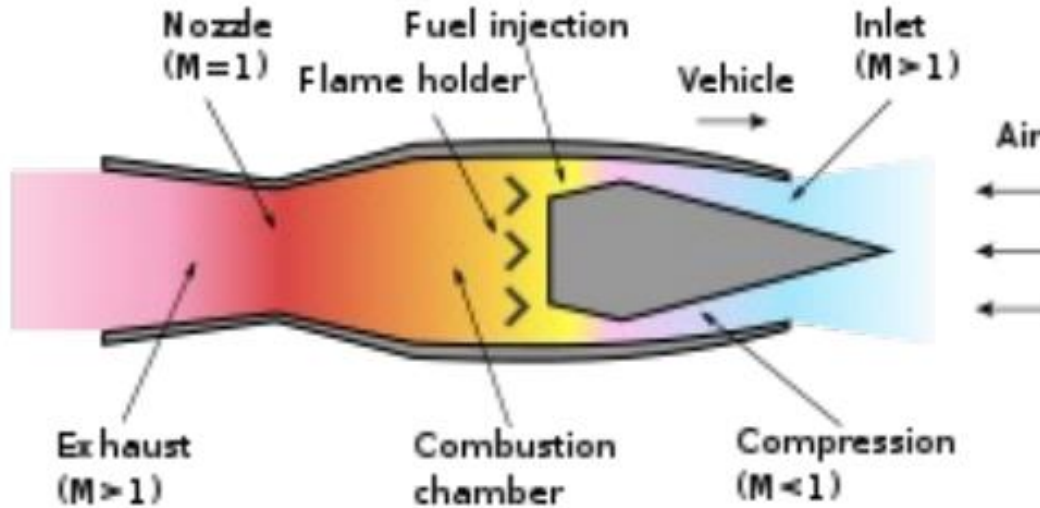
Jet Engines

- A jet engine is a type of reaction engine discharging a fast moving jet that generates thrust by jet propulsion.
- While this broad definition can include rocket, water jet, and hybrid propulsion, the term jet engine typically refers to an internal combustion airbreathing jet engine such as a turbojet, turbofan, ramjet, or pulse jet. In general, jet engines are internal combustion engines



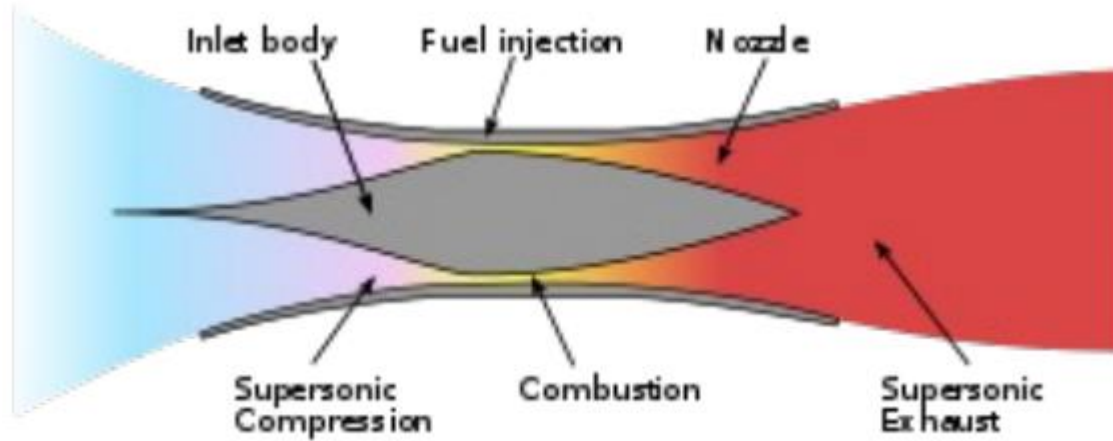
Ramjet Engine (word breakdown)

- A ramjet, or athodyd (aero thermodynamic duct), is a form of airbreathing jet engine that uses the forward motion of the engine to produce thrust. Since it produces no thrust when stationary (no ram air) ramjet-powered vehicles require an assisted take-off like a rocket assist to accelerate it to a speed where it begins to produce thrust. Ramjets work most efficiently at supersonic speeds around Mach 3 (2,300 mph; 3,700 km/h) and can operate up to speeds of Mach 6 (4,600 mph; 7,400 km/h).



Scramjet Engine

- A scramjet (supersonic combustion ramjet) is a variant of a ramjet airbreathing jet engine in which combustion takes place in supersonic airflow. As in ramjets, a scramjet relies on high vehicle speed to compress the incoming air forcefully before combustion (hence ramjet), but whereas a ramjet decelerates the air to subsonic velocities before combustion using shock cones, a scramjet has no shock cone and slows the airflow using shockwaves produced by its ignition source in place of a shock cone. This allows the scramjet to operate efficiently at extremely high speeds



SOLID FUEL DUCTED RAMJET (SFDR) TECHNOLOGY

- It is a missile propulsion system that includes a thrust modulated ducted rocket with a reduced smoke nozzle-less missile booster.
- First flight of SFDR, developed under a joint Indo-Russian R&D project, was tested in 2018. It had achieved the speed of Mach 3.
- Solid fuelled airbreathing Ramjet Engine
- Range : 70 - 340km

Dual Mode Ramjet (DMRJ)

- When a ramjet transforms into scramjet over mach 4-8 range.
- Operate both in subsonic and supersonic combustor modes.

2nd Component: Fuel (Propellant)

Solid Fuel

- Eg: HTPB : Hydroxy terminated poly- butadione (used in PSLV, GSLV)

Advantage

- Easy Storage and transfer
- All rockets use solid fuel in 1st stage as it gives more energy.

Disadvantage

- Heavy weight - Cost increase
- Non uniform beginning.

2nd Component: Fuel (Propellant)

Liquid Fuel

- UDMH (N204) (Unsymmetrical DiMethyl Hydrazine)
Used in PSLV and GSLV

Advantage

- Higher efficiency : Uniform Burning : **Demonstration**
- Less weight, So cheaper storage

Disadvantage

- Difficulty in storage (flammable/corrosive) and transport

Cryogenic Engine

Indigenous developed by India!

- Stores fuel (Liquid hydrogen) and oxidiser (Liquid O₂) at very low temp.
- H₂ = 20K. (-253 degree C)
- O₂ = 90K (-183 degree C)
- Generates High energy thrust by burning Hydrogen
- Hydrogen is
 - Abundant
 - Environment Friendly (burns to produce H₂O)

Why Cryogenic Engine?

- Hydrogen and Oxygen are stored as Liquid. Why? If they were stored as gases, it would require a huge container and the chances of leakage would be very high.
- Why Oxygen and Hydrogen? : Highest efficiency
- Provides best thrust per kilogram of fuel combination.
- Challenge of American Obstructionism due to non membership of the MTCR.